03 — No-Battery Reference World (the hand-built "twin ship")

Proves · The no-battery reference world — this scenario IS world B of 01's Battery Benefit, written down as its own file.

Mechanics this scenario turns on

- Battery Benefit runs the pipeline **twice**: world A (demand = average + uncovered `L`) and world B (demand = average + the full swing `H`). `Benefit = (FOC_B − FOC_A) × hours`. Both are **optima**; a pinned baseline is ignored. Unticking "Enable Battery" in the UI gives a *third* world — raw demand, swing carried by nobody — which burns less than A and is not the comparison.
- Baseline rule: **no battery → the worst combination** (`count − 1`); **battery active → the third from worst** (`Math.Max(0, count − 3)`). It models what the ship is assumed to do today.
- The shaft generator is filled before any auxiliary starts (the main engine is already turning), and its output is a load **on** that main engine — which is why the ME figure exceeds propulsion. SG capacity scales with the number of running MEs.

Panels described below · Battery Contribution · Power Demands · Baseline & IL1 · The proof this file exists for

Anything not described here behaves exactly as in `01-excel-baseline` — same plant, same hours; only what this scenario changes is worked through below.

Trust · verified against the reference workbook. These figures are proof.

Read after · scenario 01.

No battery; loads pre-inflated to average + FULL swing: propulsion **12 036.15** (11 463+573.15), hotel **3 876** (3 800+76). This is exactly the internal "world B" the Battery Benefit is computed against — built by hand so you can see it.

Battery Contribution

Absent — no battery configured. (Tiles in the form show nothing.)

Power Demands

SG 3 250 · AE 626 · ME = 12 036.15+3 250 = **15 286** (63.69 %) · Energy ME 15 286.15×5 000 = **76 430 750** (remember: ME energy includes driving the SG).

Baseline & IL1

No battery ⇒ baseline default = **last row** (worst: 2 ME+SG, 2.7305 → **13 652.6 t/yr**), not 3rd-from-worst — the Krishna rule applies only when a battery is active. IL1 optimal = 2.69198 → **13 459.9 t/yr**; savings 192.7 t.

The proof this file exists for

```
13 459.9  −  13 286.2 (IL1 of test 01)  =  173.7  = test 01's green badge  ✓
13 459.9  −  13 371.2 (IL1 of test 02)  =   88.7  = test 02's green badge  ✓
```

One file proves both benefits, because the no-battery twin is the same ship in both comparisons.

Takeaway: the dual-scenario (R3a) mechanism reproduced with two imports and a subtraction — no trust in hidden code required.